STATS 280 W26 - Midterm 1 - Solutions

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Ex.1 (30 points, with 2 points free)

a 7 points

The evidence does suggest that screening saves lives based on the breast cancer rate of death being 1.3 per thousand women in the treatment group and 2.0 per thousand women in the control group.

b 7 points

Screening did not cut the death rate from all causes in half. Women who accepted screening tended to be wealthier and hence less likely to die from other diseases than the women who did not accept screening.

c 7 points

Comparing the Examined group to the Refused group would be a biased comparison because women in the Examined group tended to be richer and the women in the Refused group poorer. If poorer women are more likely to die of breast cancer than richer women, then the comparison would be biased in favor of screening. (Aside: it's odd that the death rate for breast cancer is higher in the control group than in the Refused group, which has a higher percentage of poorer women than the control group. One possible explanation is that richer women are more likely to get breast cancer than poorer women. This is potentially due to lifestyle choices, such as having children later, having fewer children, and increased alcohol consumption. The increased death rate from breast cancer in the control group could be driven by wealthier women who do not get breast cancer diagnoses until the cancer has progressed past the point where it is treatable.)

d 7 points

You expect to find more cancers per thousand women in the Examined group because you are screening for cancer in the Examined group, as opposed to the control group. Screening does not cause breast cancer.

Ex.2 (30 points)

a 8 points

The average GPA in group 1 is a weighted average of the GPAs of the math and statistics majors, which is a weighted average of 3.2 and 3.3. This will be at least 3.2. The average GPA in group 2 will be a weighted average of 3.0 and 3.1, which will be at most 3.1. Hence the average GPA in group 1 is higher than the average GPA in group 2.

b 8 points

It is possible here for the average GPA in group 2 to be higher than the average GPA in group 1 if most students in group 2 are statistics majors while most students in group 1 are math majors. Suppose that 49 students in group 1 are math majors and 1 student a statistics major, while 1 student in group 2 is a math

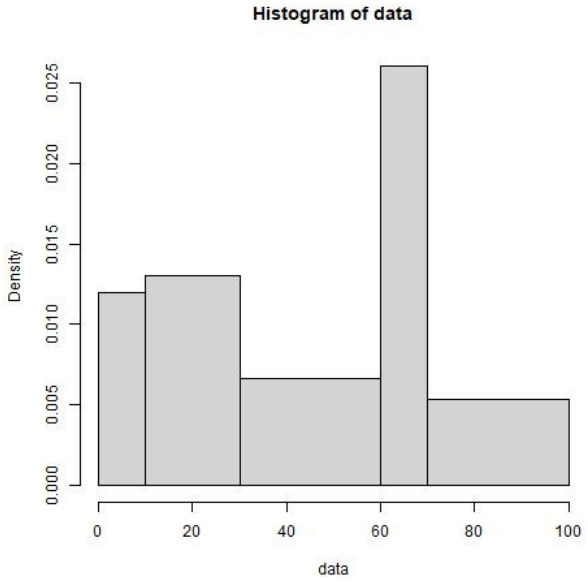


Figure 1: Histogram for 3a

major and 49 students are statistics majors. The average GPA in group 1 will be $(49*3.2+1*3.3)/50 = 3.20$ while the average GPA in group 2 will be $(1*3.0+49*3.25)/50 = 3.25$. This is an example of Simpson's paradox.

c 7 points

This is an observational study since students are not randomly enrolled into private or public high schools.

d 7 points

The difference in admission percentages could be explained by differences in socioeconomic status or extracurricular college preparation between private and public high school students.

Ex. 3 (35 points)

a 15 points

Range	Count	Percentage	Density
$[0, 10)$	6	$6/50 = 0.12$	$0.12/10 = 0.012$
$[10, 30)$	13	$13/50 = 0.26$	$0.26/20 = 0.013$
$[30, 60)$	10	$10/50 = 0.2$	$0.2/30 = 0.0067$
$[60, 70)$	13	$13/50 = 0.26$	$0.26/10 = 0.026$
$[70, 100)$	8	$8/50 = 0.16$	$0.16/30 = 0.0053$

b 10 points

The assumption that the numbers are evenly distributed in each interval allows us to assume that all of the values in each block are centered in the middle of the block for the purposes of estimating the mean. Thus we can estimate the mean as $5*0.12+20*0.26+45*0.2+65*0.26+85*0.16 = 45.3$.

We know that the median must be in the block $[30, 60)$ since 38% of the values are less than 30 and 58% are less than 60. If x is the median, then 12% of the values are between 30 and x. If the values are evenly distributed in the block $[30, 60)$, then we would estimate x as $30 + (12/20) * (60 - 30) = 48$.

c 5 points

You should guess the median 48.

d 5 points

Twice the number you guess should be the mean 45.3. Hence you should guess $45.3/2 = 22.65$.

Ex.4 (35 points)

a 7 points

This is the percentage within 2 standard deviation of the mean, which is roughly 95%.

b 7 points

Converting to standard units gives $(49.9 - 50)/0.2 = -0.5$ and $(50.2 - 50)/0.2 = 1$. Roughly $38.3\%/2 + 68.3\%/2 = 53\%$ of rods will be in this range.

c 7 points

The 80th percentile corresponds to 60% of the area in the middle, which corresponds roughly to $z = 0.84$. Hence the length of the rod will be $50 + 0.84 * 0.2 = 50.17$ cm.

d 7 points

Markov's inequality gives a maximum percentage of $\$60,000/\$100,000 = 60\%$. This would occur when 60% of citizens earn $\$100,000$ per year and 40% of citizens earn \$0 per year.

e 7 points

Converting to standard units gives $(\$15,000 - \$60,000)/\$15,000 = -3$ and $(\$105,000 - \$60,000)/\$15,000 = 3$. Chebyshev's inequality gives a maximum of $1/3^2 = 1/9 = 11\%$.

Ex.5 (30 points)

a (7 points)

If the correlation is 1, then all of the points would lie on a straight line. This cannot happen, since the slope between $(1, 6)$ and $(3, 14)$ is $(14-6)/(3-1) = 4$ while the slope between $(3, 14)$ and $(5, 26)$ is $(26-14)/(5-3) = 6$. Hence there is no value x_2 that will make the correlation 1.

b (8 points)

The covariance must be 0 for the correlation to be 0. Hence:

$$\begin{aligned} 0 = cov =& \overline{xy} - \bar{x}\bar{y} \\ =& (6 + 10x_2 + 42 + 130)/4 - (1 + x_2 + 3 + 5)/4 * (6 + 10 + 14 + 26)/4 \\ =& (10x_2 + 178)/4 - (x_2 + 9)/4 * 56/4 = (10x_2 + 178 - 14(x_2 + 9))/4 \\ =& (-4x_2 + 178 - 126)/4 = (-4x_2 + 52)/4 = -x_2 + 13. \end{aligned} \tag{1}$$

Hence $x_2 = 13$.

c (15 points)

From left to right, $0.1, 0.7, -0.3$.

Ex.6 (40 points)

a 8 points

The 3.2 cups per day corresponds to $z = (3.2 - 2)/1 = 1.2$. The regression method would predict $z = 1.2 * 0.2 = 0.24$ for steps per day, which corresponds to $6,000 + 0.24 * 1200 = 6,288$ steps per day.

b 8 points

The slope of the line will be $0.2 * 1200/1 = 240$ and the line will go through the point of averages $(2, 6,000)$. Hence the equation of the line will be $y - 6,000 = 240(x - 2)$ or $y = 240x + 5,520$.

c 12 points

The average steps per day for someone who drinks 3.2 cups of coffee per day will be $6,288$ from part a), and we can estimate the standard deviation in the vertical strip as the rms error $\sqrt{1 - 0.2^2} * 1200 = 1175.76$. The 10th percentile corresponds to 80% of the area in the middle, which gives $z = -1.28$ and the 90th percentile corresponds to $z = 1.28$. Hence the 10th percentile will be roughly $6,288 - 1.28 * 1175.76 = 4,783$ steps per day and the 90th percentile $6,288 + 1.28 * 1175.76 = 7,793$ steps per day.

d 12 points

The 80th percentile corresponds to $z = 0.84$ from problem 4c). We would predict $z = 0.2 * 0.84 = 0.168$ for steps per day, or $6,000 + 0.168 * 1200 = 6,201.6$ steps per day. Using the rms error from part c), we get $8,000$ steps per day corresponding to $z = (8,000 - 6,201.6)/1,175.76 = 1.53$. The right tail will be roughly $(100\% - 87.4\%)/2 = 6.3\%$.